

NASM | CPT7_Ch13_Integrated Training and the OPT Model

Hey, everybody. Welcome to your Chapter 13 review. In this episode, we'll talk about integrated training and the optimum performance training, or OPT model.

Nothing beats having a structured, science-based, proven and sequential program as your foundation. NASM gives you that. The backing and credibility that comes from your certification is what's going to take you to the next level, and give you a long and fulfilling career as a fitness professional.

We know the best way to achieve consistent, superior results is to follow a structured training program like the NASM OPT model. It's evidence-based, progressive and intuitive. You're going to love it. Flexibility, cardiorespiratory, core, balance, plyometric, speed, agility, quickness, or SAQ, and resistance training all come together in one system to create the OPT models integrated training concept. It's an all-inclusive approach to exercise that can lead to improvements in overall health, wellness and athletic performance.

Not every client will need, or even want, every single component. It's like they describe in the textbook-- you can have a giant chef's salad, but you might leave out certain ingredients for certain people, or add extras, based on what they need and prefer. Using integrated training principles, you'll have the ability to provide a more comprehensive approach to program design and personal training.

You'll utilize an array of training strategies that are unified and tailored to specific health and fitness goals, and likely even discover new and exciting exercises, movements, and variables to add to your professional toolbox. Leveraging these integrated training concepts when selecting exercises and designing programs will help you plan and execute a client-centered approach to fitness training. Your clients won't be just working out, they'll be getting psychological, physiological, body composition, and performance benefits.

Upon completion of this chapter, you should be able to summarize integrating training principles and their importance to maximize function, identify types of exercise within an integrated training model, list the five phases of NASM's OPT model, and describe the benefits of using optimum performance training with clients. When we say it's important to start with a systematic approach, we mean ensuring solidity and stability. You wouldn't build a house without first laying a foundation, right? The same concept applies to training. The body must also have a solid foundation of fitness before embarking on a more intense training regimen in order to maintain safety and effectiveness.

Now, with the myriad of choices and modifications, most exercises involve at least one of several standard movement patterns, and others may combine multiple movement patterns into a single exercise. Your basic movement patterns are the squat, hip hinge, pulling, and pressing motions. You need to understand the functions and benefits of each one, so you can expand off of them to personalize workouts that fit your clients' needs.

One primary goal with any and all of these moves is to maintain, or correct, posture and range of motion, or ROM for short. Also, and this is important, exercise programs should involve training in all three planes of motion-- sagittal, frontal and transverse-- in order to enhance muscle recruitment and challenge the body functionally. Apply acute training variables by manipulating things like repetitions, sets, intensity, tempo or speeds of movement, rest intervals, exercise selection, exercise order, as well as training volume, frequency, and duration. They really are what ultimately determines how a client's body responds to an exercise program.

Arguably, exercises can be distilled down to these individual variables. The same exercises incorporating different variables will elicit different results. Your program should also correlate with the assessment results taken prior to the initial workout. Let's dig a little deeper into each component.

First up, flexibility training. This increases joint ROM, and involves different types of stretching techniques to improve tissue extensibility. NASM recommends using a variety of stretching techniques, including self-myofascial techniques such as foam rolling, as well as static, active, and dynamic stretching. Aside from the benefits of improving ROM, you'll also be helping avoid sore muscles and potential injuries.

Cardiorespiratory training refers to the capacity of the respiratory and cardiovascular systems to provide muscles with oxygen during sustained and/or intense exercise. Typical forms of cardiorespiratory exercise include walking, jogging, running, cycling, biking, swimming, rowing, and just about any sport you can think of that requires some sustained bouts of exertion. Other benefits of cardio training include decreased resting heart rate and blood pressure, and increased stroke volume and cardiac output. Those things are hugely important for health and longevity.

Core training is a biggie because of its ability to enhance posture, which is, in a lot of ways, the key to a healthy kinetic chain. Core training strengthens muscles of the abdomen, lower back, and pelvis, and can minimize low back pain. In addition, core training improves athletic ability and rotational power of the torso.

A lot of people equate core work to a smoking six pack. But really, the benefits of core training are more about better posture and enhanced bodily function for daily living. Increased balance, stabilization, and coordination of the kinetic chain is something we take for granted, until we throw something out of whack. Having appropriate core stability and strength can help avoid that. And hey, if your client wants a six pack, more power to them.

Next up is balance training. Balance training is a method that can create better movement patterns for all clients, reduce the occurrence of falls in the elderly, and ankle sprains in athletes. Balance training also improves proprioception, and agility-based activities. I can tell you from personal experience, this will come in handy the next time you get soap in your eyes while showering, knock a bottle of shampoo off a shelf, and have to react quickly before it lands on your big toe. Try doing that without well-honed proprioception, balance, and agility skills.

Moving on to plyometric training, also known as reactive or jump training. Plyometric exercise improves the stretch shortening cycle, which enhances a client's ability to move explosively. The benefits of plyometric training even include improved bone mineral density and soft tissue strength. It also increases metabolic demands and caloric expenditure, which is great for your clients with weight management goals who are capable of these types of movement.

Next, speed, agility, and quickness, or SAQ training, enhances the capability of individuals to be more skillful at faster speeds and with greater precision. SAQ training aims to improve acceleration, deceleration, and change of direction abilities. These are also great exercises for burning calories. You'll also find that your performance athletes will get a lot of benefits from this area.

Our final component, resistance training, can promote marked increases in muscular endurance, strength, and hypertrophy, regardless of age or gender. Resistance training plays a fundamental role in the improvement of athletic performance and rehabilitation from musculoskeletal injury. It's also an effective method for increasing lean body mass and reducing body fat.

That's a pretty high level overview. You'll want to read up on each component on your own to really gain a more thorough understanding. Let's move on to explore the OPT model. The OPT model is based on the science of human movement, and uses the principle of specificity and integrated training to maximize client results. It provides a system for safely and effectively progressing a client toward their goals by using a sequential training method consisting of three levels-- stabilization, strength, and power.

The stabilization level has one phase of training-- stabilization endurance training. It is designed to teach optimal movement patterns for joint and core stabilization and postural control. It focuses on exercises and movements that enhance muscular endurance, joint ROM, joint and core stability, and neuromuscular coordination. This is done through slow and controlled movements with relatively light loads, and meticulous attention to both posture and exercise technique. I think it's safe to say you'll utilize this level and phase for everyone.

The strength level consists of Phases 2, 3, and 4. The goal of Phase 2, strength endurance training, is to enhance stabilization endurance while increasing prime mover strength. These two adaptations are accomplished by performing two exercises with similar joint dynamics in a superset sequence, meaning back-to-back without rest.

Phase 3, muscular development training, is designed for folks who are looking to maximize muscle growth, or hypertrophy, like bodybuilders and others interested in physique development, individuals seeking body composition change, as well as certain athletes. This phase of training can use a variety of resistance exercises and methods to elicit responses in muscle growth, which can also be beneficial for clients who want to lose body fat while maintaining lean muscle mass.

Phase 4, maximal strength training, works toward the goal of maximal prime mover strength by lifting heavy loads. This is going to be more common for strength athletes and others who require higher levels of strength, so this is a phase you may not even get to with some individuals. It is an advanced form of exercise that is optional, depending on the goals of the client.

Within the power level of the OPT model, only one phase of training is covered. This is Phase 5, otherwise known as power training. This is where you really max things out only after a client successfully completes the stabilization and strength levels, or if they're already exceptionally well conditioned, exhibit adequate joint and core stability, and are familiar with PowerForms of exercise.

The goal of this phase is to increase maximal strength and rate of force production. Similar to Phase 2, this is accomplished by performing a superset sequence of two exercises with similar joint dynamics. Only, in this phase, the first exercise will be focused on challenging max strength, and the second on challenging velocity.

As you can see, the OPT model isn't necessarily intended to expose every client to every exercise and move them through each and every one of the phases. Sometimes it's all about how you use exercise variations to get the body to respond in a different way. The OPT model provides a structured way to take every one of your clients to the level they want and need to go. With all these tools at your disposal, you can create a plan that makes sense, builds confidence, works effectively, keeps things interesting, and minimizes injury.

All right, that's going to do it for this episode. Take care. And happy studying, everybody.